

TIO-01-System Architecture

Overview:

This is the first lecture of the “Trading Tech, Infra & Operations” module where the instructor explains the evolution of automated trading system platforms - different components of automated trading systems and how they link with each other (i.e., how they fit into the bigger picture). The session focuses on how quantitative traders can fine-tune their automated systems for the specific style of their quantitative trading strategies - optimally leveraging the power of technology by optimizing the different components suited to their requirements.

To be able to appreciate the power of technology in automated trading systems, the instructor first explains the system architecture of a traditional trading system and then explains the architecture of a modern-day automated trading system - so that participants can appreciate the innovations that have happened and have been added over the years. The instructor also explains the key components of both these trading systems; and discusses the challenges that need to be handled/overcome to build robust automated trading systems.

Key Takeaways:

- Understanding different components of a traditional trading system.
- Understanding different components of an automated trading system.
- Understanding how different components of an automated trading system are linked to each other, and the challenges involved in building a trading system.

Pre-reading/Pre-lecture tasks:

- Look up brochures of different automated trading systems on the Internet
- Go through the [Glossary](#) to understand the basic terms used in trading

Use of the topics learned in this session:

Post this lecture, you will be in a position to understand:

- The major components of an automated trading system. This will be especially useful when leading a trading desk or setting up your trading desk. As a trader you will need to optimize each component as per the needs of your strategy - since there is no single optimal design for all types of trading strategies, the understanding acquired in this session can be potentially helpful.
- The entire cycle of data flow - from the Exchange to the Server, the processing with the Server (including storage, complex event processing, and order generation), and the flow of orders from the Server to the Exchange.
- Different components of a trader's application and risk management.
- The importance of latency between the origins of the event to the order generation to beat the competition.

- Detailed understanding of the Complex Event Processing engine.
- Awareness about technological protocols (and standardized protocols like FIX protocol) used by the exchange to communicate with members.

Additional read:

Blog: [Decoding the Black Box running Trading Systems](#)

Recommended time for the session & related coursework - 7 hours